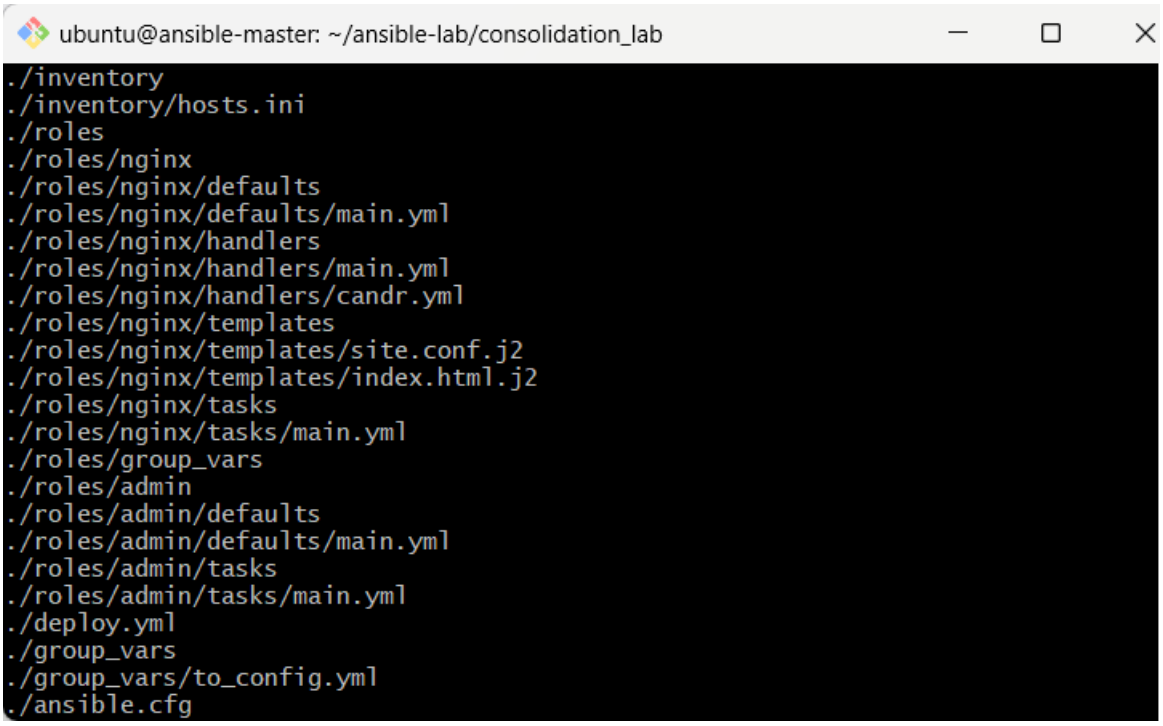
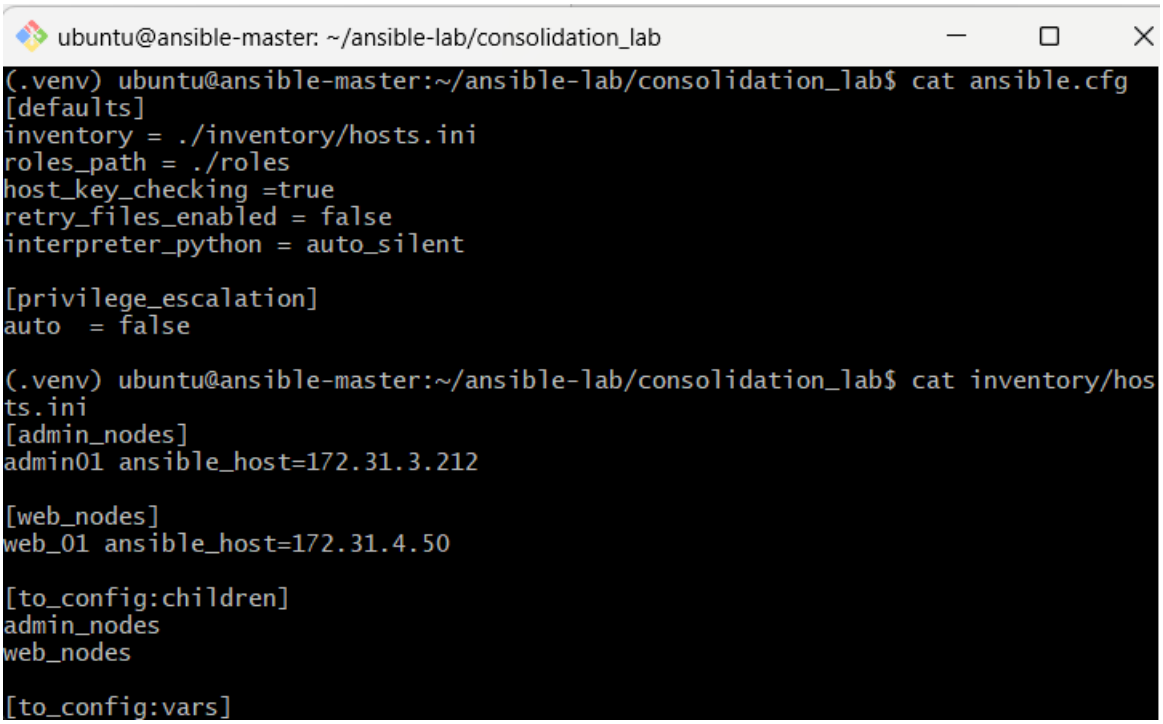
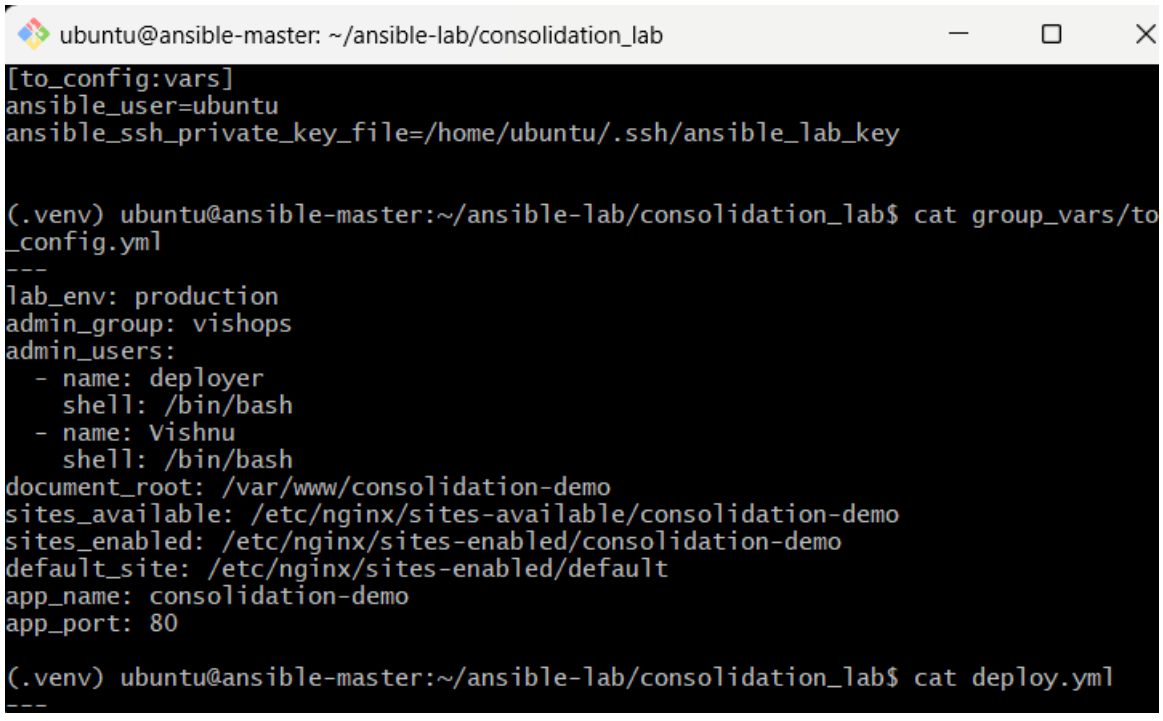
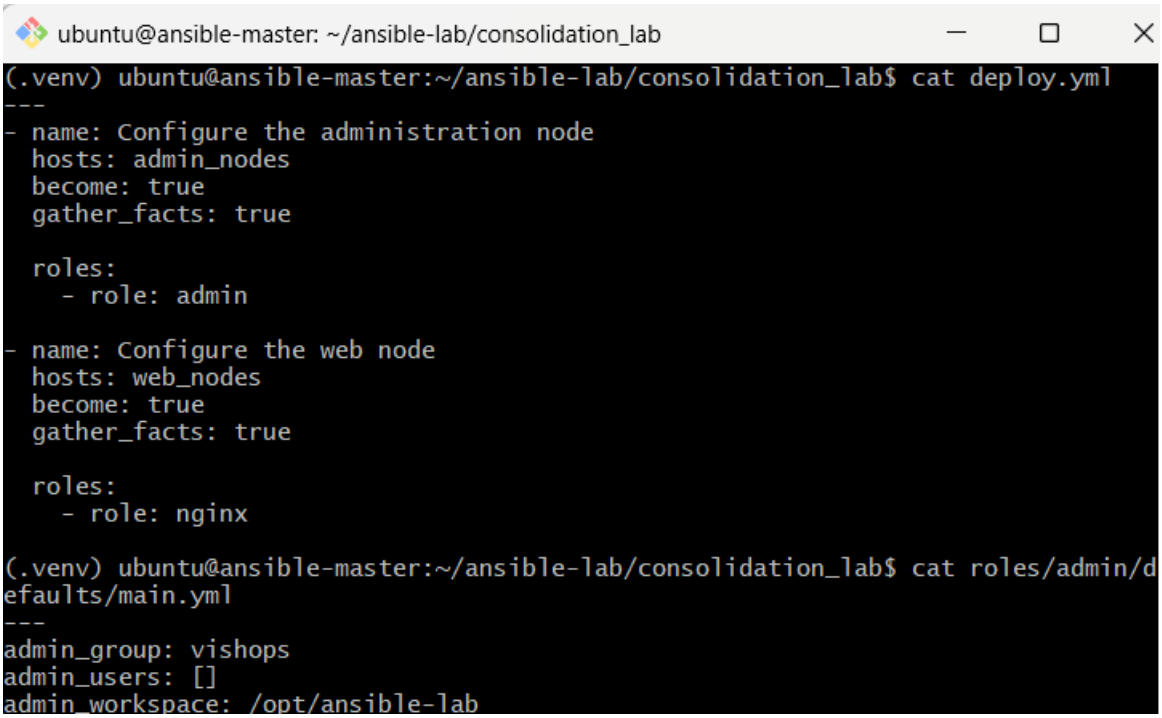
Ansible Mastery Part-7

Advanced Configuration Automation Across Managed Nodes: Loop-Driven User Provisioning, Conditional Task Execution, Environment Variable Management, and Templated Nginx Deployment with Declarative Site Configuration, Event-Driven Handlers, and Automated HTTP Health Verification




ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat ansible.cfg
[defaults]
inventory = ./inventory/hosts.ini
roles_path = ./roles
host_key_checking =true
retry_files_enabled = false
interpreter_python = auto_silent
[privilege_escalation]
auto = false
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat inventory/hos
ts.ini
[admin_nodes]
admin01 ansible_host=172.31.3.212
[web_nodes]
web_01 ansible_host=172.31.4.50
[to_config:children]
admin_nodes
web_nodes
[to_config:vars]


ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
[to_config:vars]
ansible_user=ubuntu
ansible_ssh_private_key_file=/home/ubuntu/.ssh/ansible_lab_key
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat group_vars/to
_config.yml
---
lab_env: production
admin_group: vishops
admin_users:
  - name: deployer
    shell: /bin/bash
  - name: Vishnu
    shell: /bin/bash
document_root: /var/www/consolidation-demo
sites_available: /etc/nginx/sites-available/consolidation-demo
sites_enabled: /etc/nginx/sites-enabled/consolidation-demo
default_site: /etc/nginx/sites-enabled/default
app_name: consolidation-demo
app_port: 80
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat deploy.yml
---

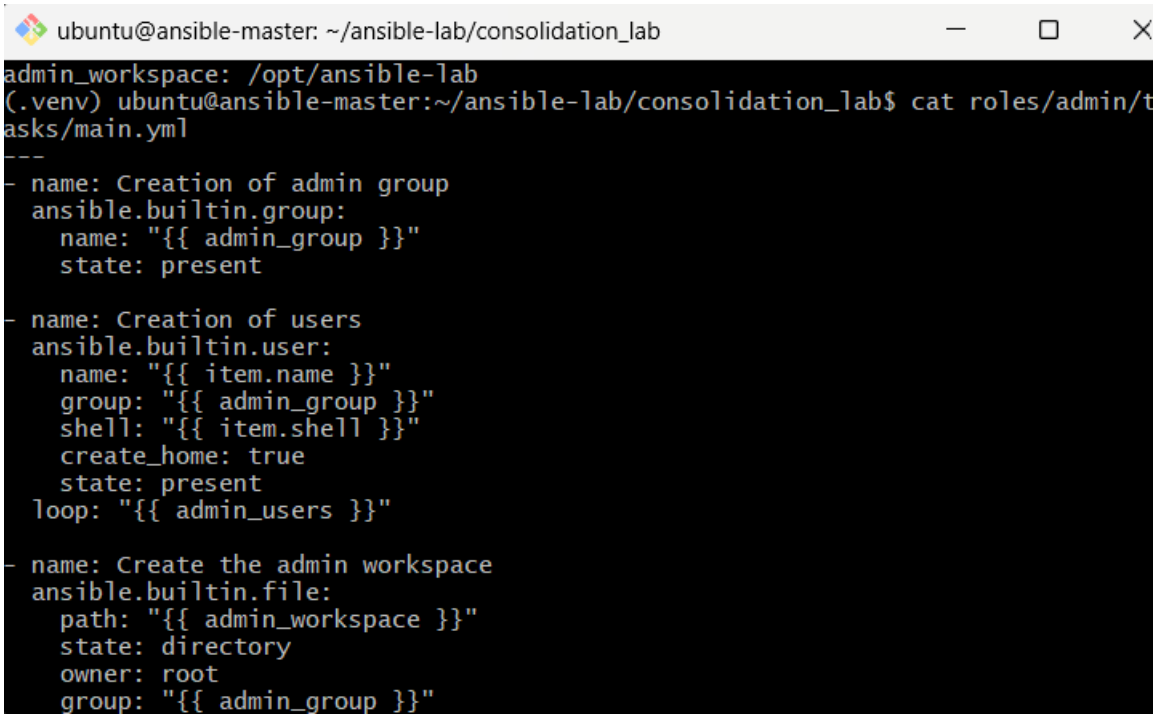

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat deploy.yml
---
- name: Configure the administration node
  hosts: admin_nodes
  become: true
  gather_facts: true

  roles:
    - role: admin

- name: Configure the web node
  hosts: web_nodes
  become: true
  gather_facts: true

  roles:
    - role: nginx

(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/admin/d
efaults/main.yml
---
admin_group: vishops
admin_users: []
admin_workspace: /opt/ansible-lab


ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
admin_workspace: /opt/ansible-lab
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/admin/t
asks/main.yml

- name: Creation of admin group
  ansible.builtin.group:
    name: "{{ admin_group }}"
    state: present

- name: Creation of users
  ansible.builtin.user:
    name: "{{ item.name }}"
    group: "{{ admin_group }}"
    shell: "{{ item.shell }}"
    create_home: true
    state: present
  loop: "{{ admin_users }}"

- name: Create the admin workspace
  ansible.builtin.file:
    path: "{{ admin_workspace }}"
    state: directory
    owner: root
    group: "{{ admin_group }}"

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab

```
    path: "{{ admin_workspace }}"
    state: directory
    owner: root
    group: "{{ admin_group }}"
    mode: "2775"

- name: Create an env specific marker
  ansible.builtin.copy:
    content: "Environment: {{ lab_env }}\n"
    dest: "{{ admin_workspace }}/environment.txt"
    owner: root
    group: "{{ admin_group }}"
    mode: "0640"

- name: Check if the env is set to production
  ansible.builtin.debug:
    msg: "Environment is set to Production"
  when: lab_env == "production"

- name: List the configured users
  ansible.builtin.debug:
    msg: "User {{ item.name }} configured with shell {{ item.shell }}"
  loop: "{{ admin_users }}"
```

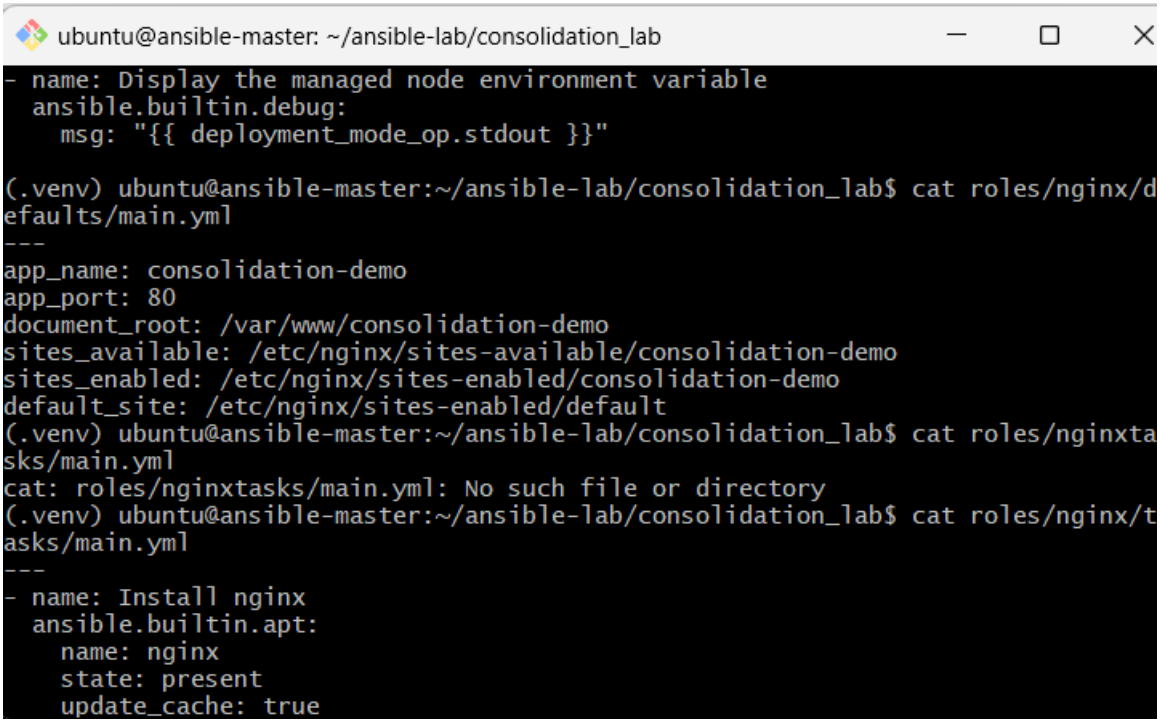
ubuntu@ansible-master: ~/ansible-lab/consolidation_lab

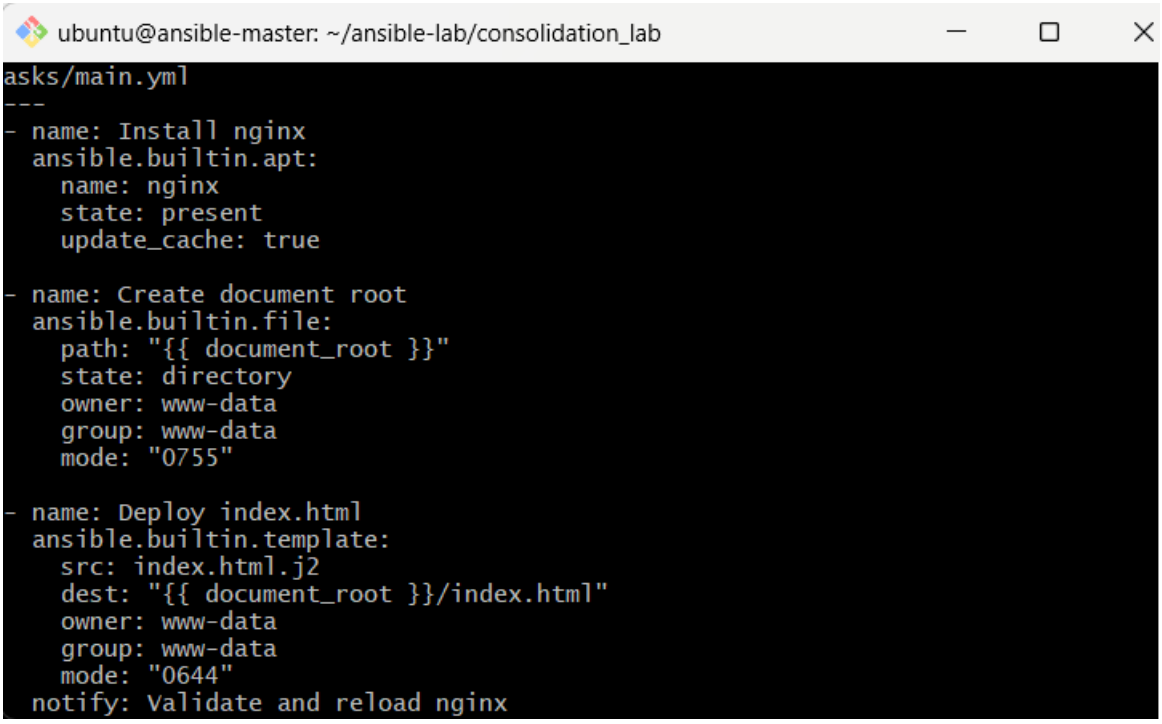
```
- name: List the configured users
  ansible.builtin.debug:
    msg: "User {{ item.name }} configured with shell {{ item.shell }}"
  loop: "{{ admin_users }}"

- name: Read a variable from the control node's env
  ansible.builtin.debug:
    msg: "Deployment label: {{ lookup('ansible.builtin.env','deployment_label',d
efault='local-lab') }}"

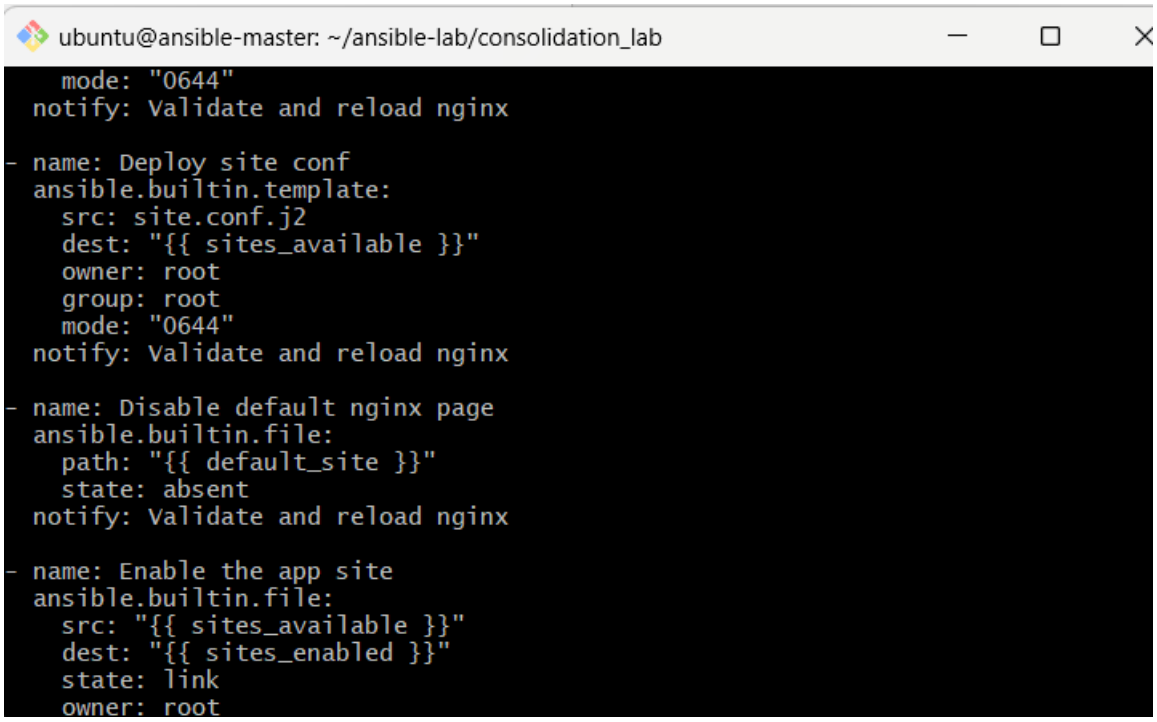
- name: Run a variable with task-specific env vars
  ansible.builtin.command:
    argv:
      - /usr/bin/printenv
      - DEPLOYMENT_MODE
  environment:
    DEPLOYMENT_MODE: "{{ lab_env }}"
  register: deployment_mode_op
  changed_when: false

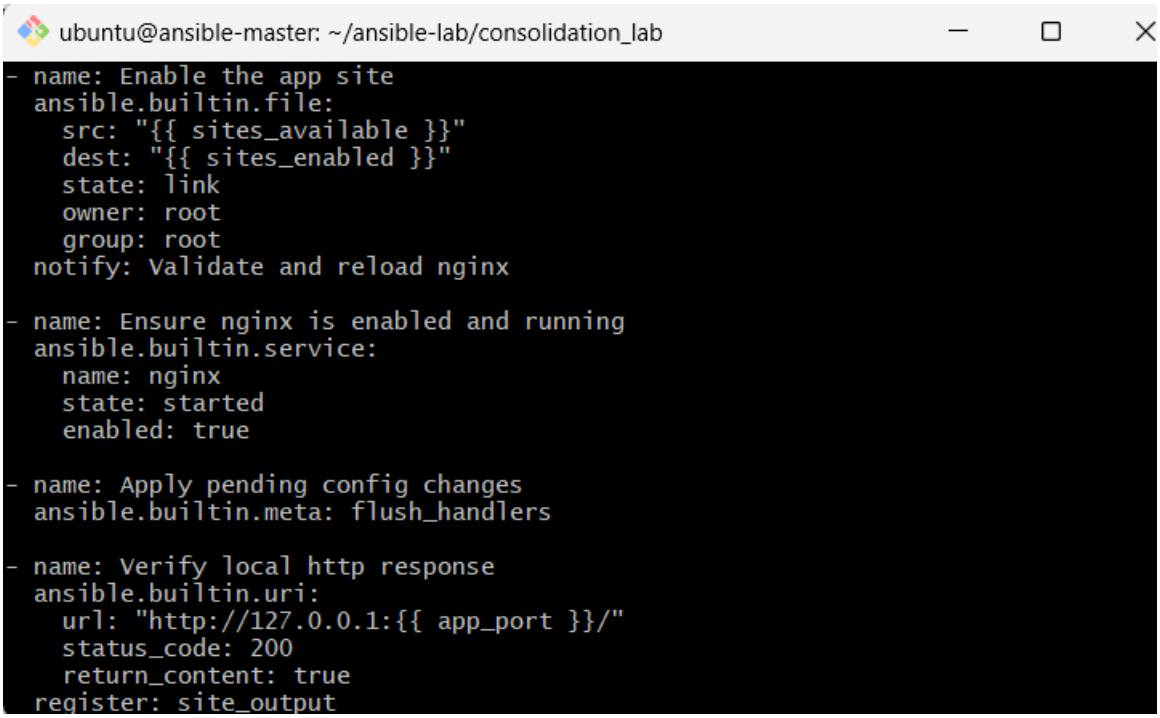
- name: Display the managed node environment variable
  ansible.builtin.debug:
    msg: "{{ deployment_mode_op.stdout }}"
```

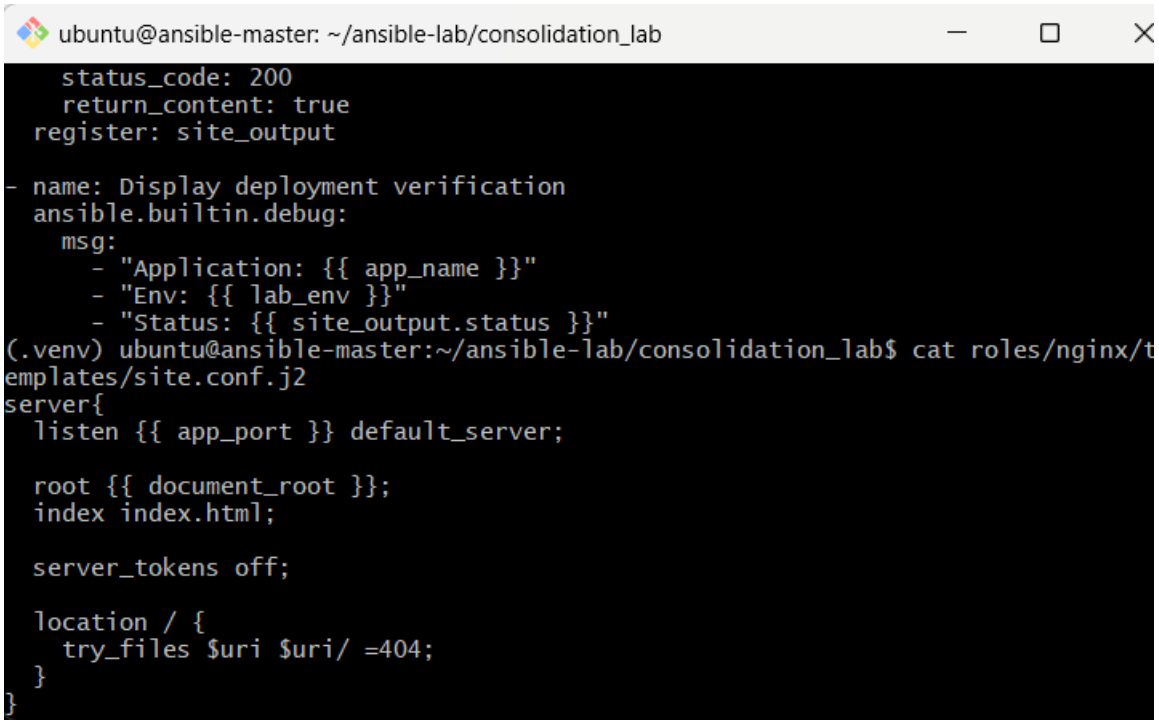

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
- name: Display the managed node environment variable
ansible.builtin.debug:
msg: "{{ deployment_mode_op.stdout }}"
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/nginx/defaults/main.yml
---
app_name: consolidation-demo
app_port: 80
document_root: /var/www/consolidation-demo
sites_available: /etc/nginx/sites-available/consolidation-demo
sites_enabled: /etc/nginx/sites-enabled/consolidation-demo
default_site: /etc/nginx/sites-enabled/default
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/nginxtasks/main.yml
cat: roles/nginxtasks/main.yml: No such file or directory
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/nginx/tasks/main.yml
---
- name: Install nginx
ansible.builtin.apt:
name: nginx
state: present
update_cache: true

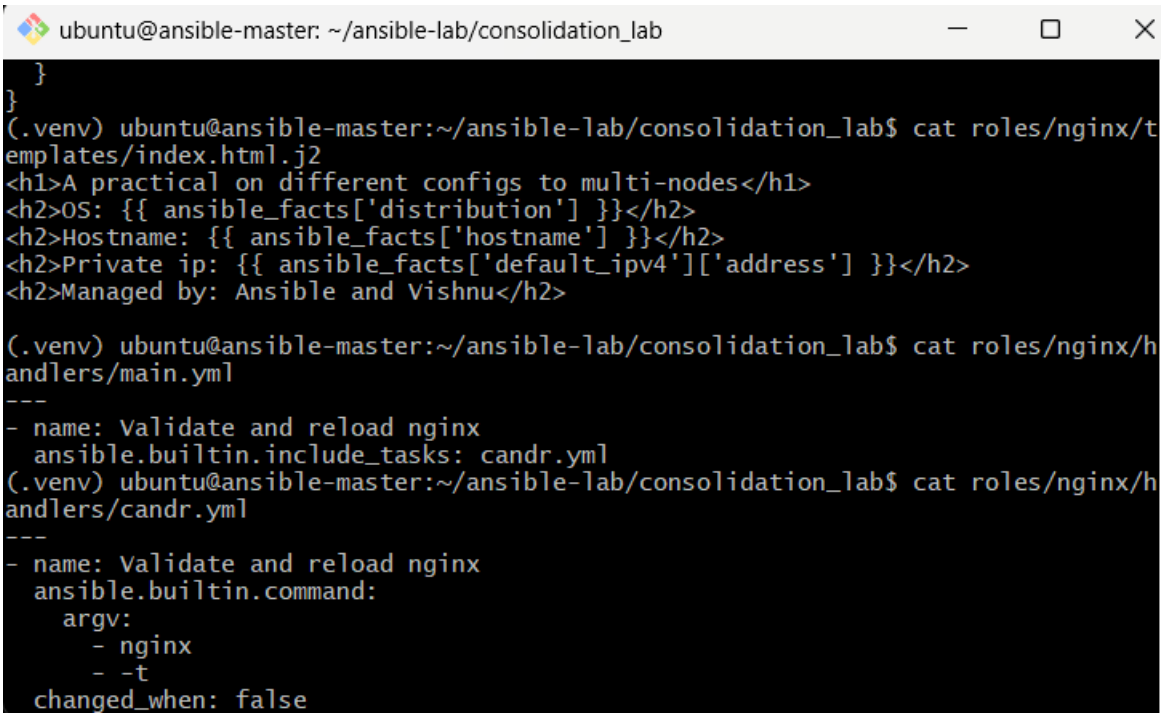

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
asks/main.yml

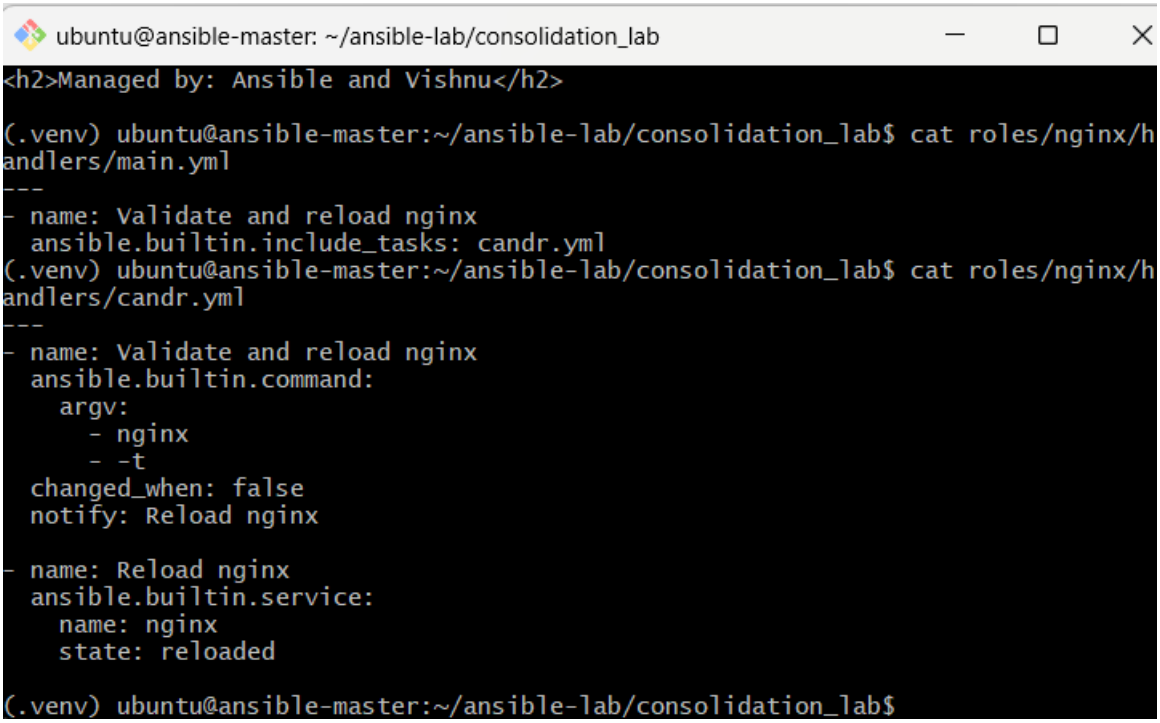
- name: Install nginx
ansible.builtin.apt:
name: nginx
state: present
update_cache: true
- name: Create document root
ansible.builtin.file:
path: "{{ document_root }}"
state: directory
owner: www-data
group: www-data
mode: "0755"
- name: Deploy index.html
ansible.builtin.template:
src: index.html.j2
dest: "{{ document_root }}/index.html"
owner: www-data
group: www-data
mode: "0644"
notify: Validate and reload nginx


ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
mode: "0644"
notify: Validate and reload nginx
- name: Deploy site conf
ansible.builtin.template:
src: site.conf.j2
dest: "{{ sites_available }}"
owner: root
group: root
mode: "0644"
notify: Validate and reload nginx
- name: Disable default nginx page
ansible.builtin.file:
path: "{{ default_site }}"
state: absent
notify: Validate and reload nginx
- name: Enable the app site
ansible.builtin.file:
src: "{{ sites_available }}"
dest: "{{ sites_enabled }}"
state: link
owner: root

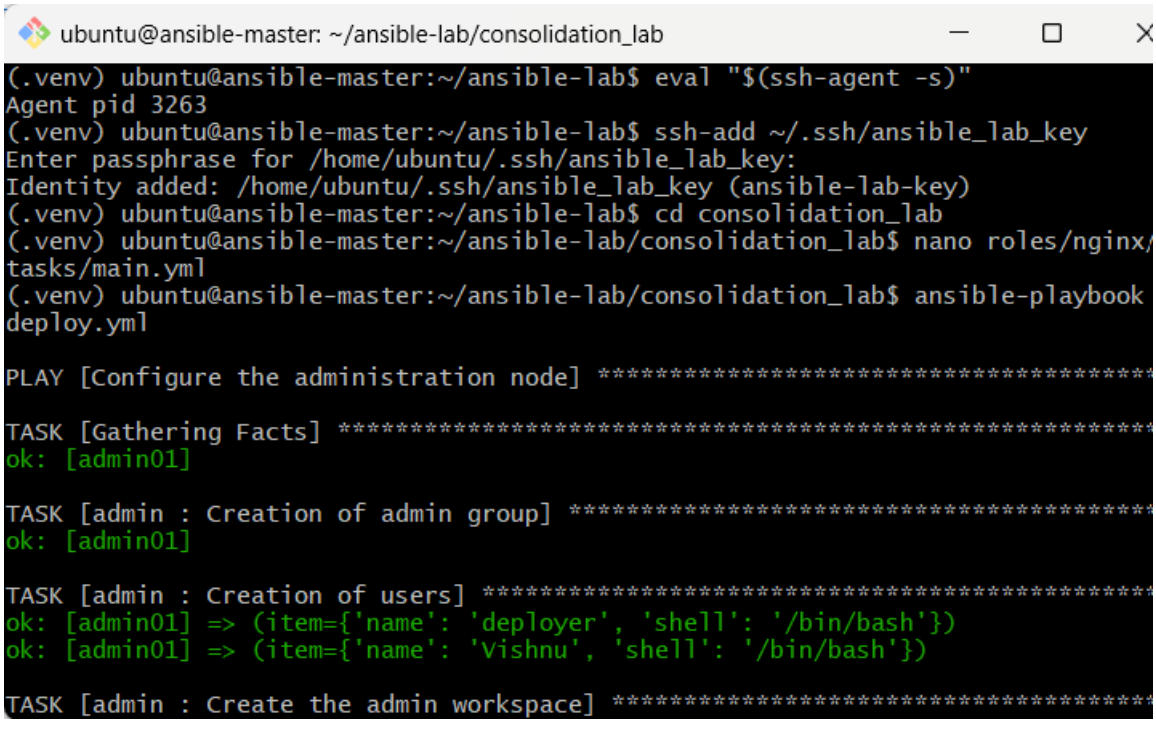

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
- name: Enable the app site
ansible.builtin.file:
src: "{{ sites_available }}"
dest: "{{ sites_enabled }}"
state: link
owner: root
group: root
notify: Validate and reload nginx
- name: Ensure nginx is enabled and running
ansible.builtin.service:
name: nginx
state: started
enabled: true
- name: Apply pending config changes
ansible.builtin.meta: flush_handlers
- name: Verify local http response
ansible.builtin.uri:
url: "http://127.0.0.1:{{ app_port }}/"
status_code: 200
return_content: true
register: site_output

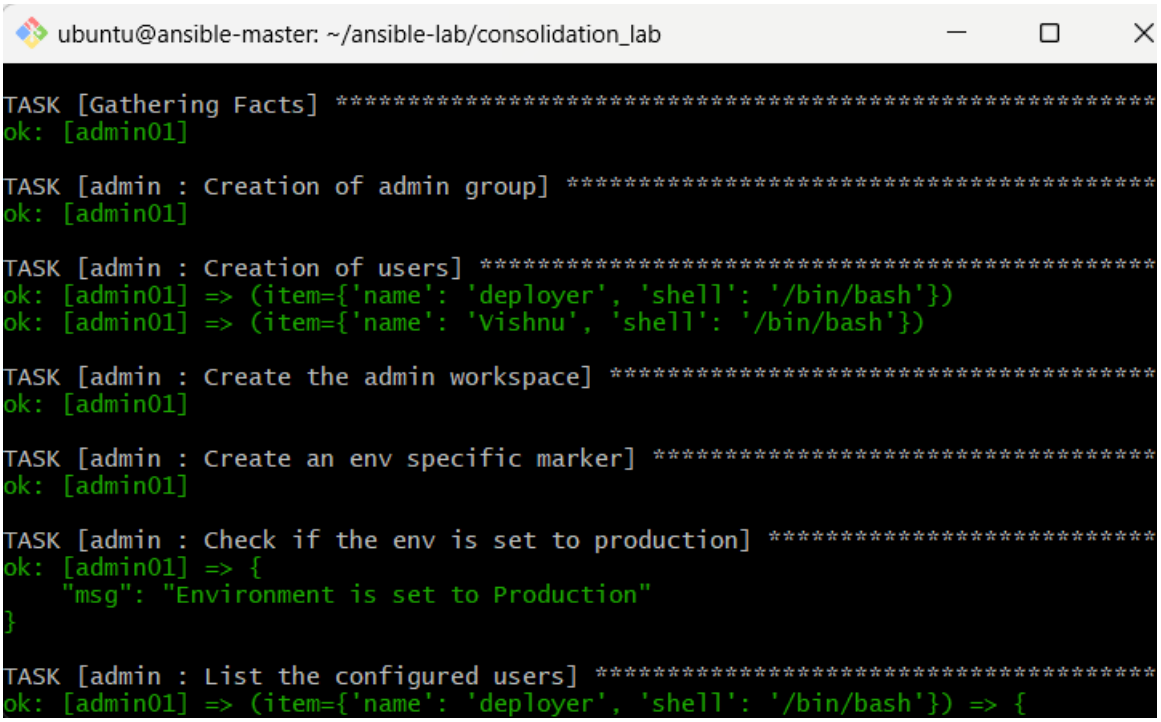

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
status_code: 200
return_content: true
register: site_output
- name: Display deployment verification
ansible.builtin.debug:
msg:
- "Application: {{ app_name }}"
- "Env: {{ lab_env }}"
- "Status: {{ site_output.status }}"
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/nginx/templates/site.conf.j2
server{
listen {{ app_port }} default_server;
root {{ document_root }};
index index.html;
server_tokens off;
location / {
try_files $uri $uri/ =404;
}
}

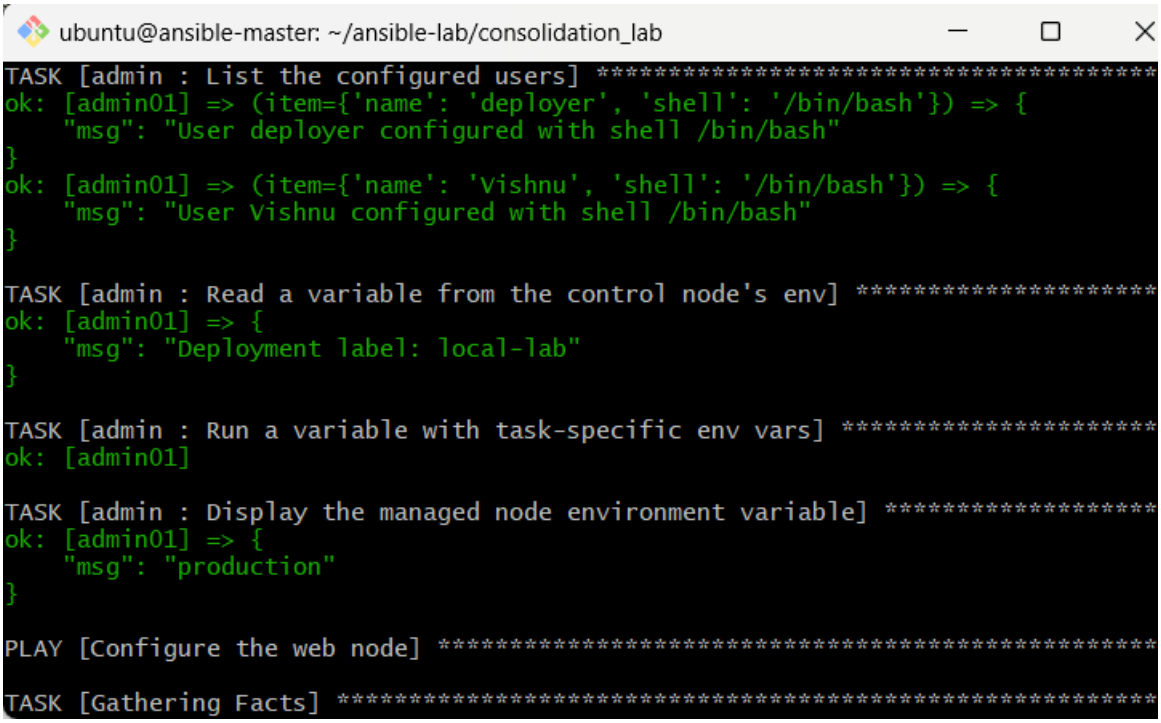

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
}
}
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/nginx/templates/index.html.j2
<h1>A practical on different configs to multi-nodes</h1>
<h2>OS: {{ ansible_facts['distribution'] }}</h2>
<h2>Hostname: {{ ansible_facts['hostname'] }}</h2>
<h2>Private ip: {{ ansible_facts['default_ipv4']['address'] }}</h2>
<h2>Managed by: Ansible and Vishnu</h2>
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/nginx/handlers/main.yml
---
- name: Validate and reload nginx
ansible.builtin.include_tasks: candr.yml
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ cat roles/nginx/handlers/candr.yml
---
- name: Validate and reload nginx
ansible.builtin.command:
argv:
- nginx
- -t
changed_when: false

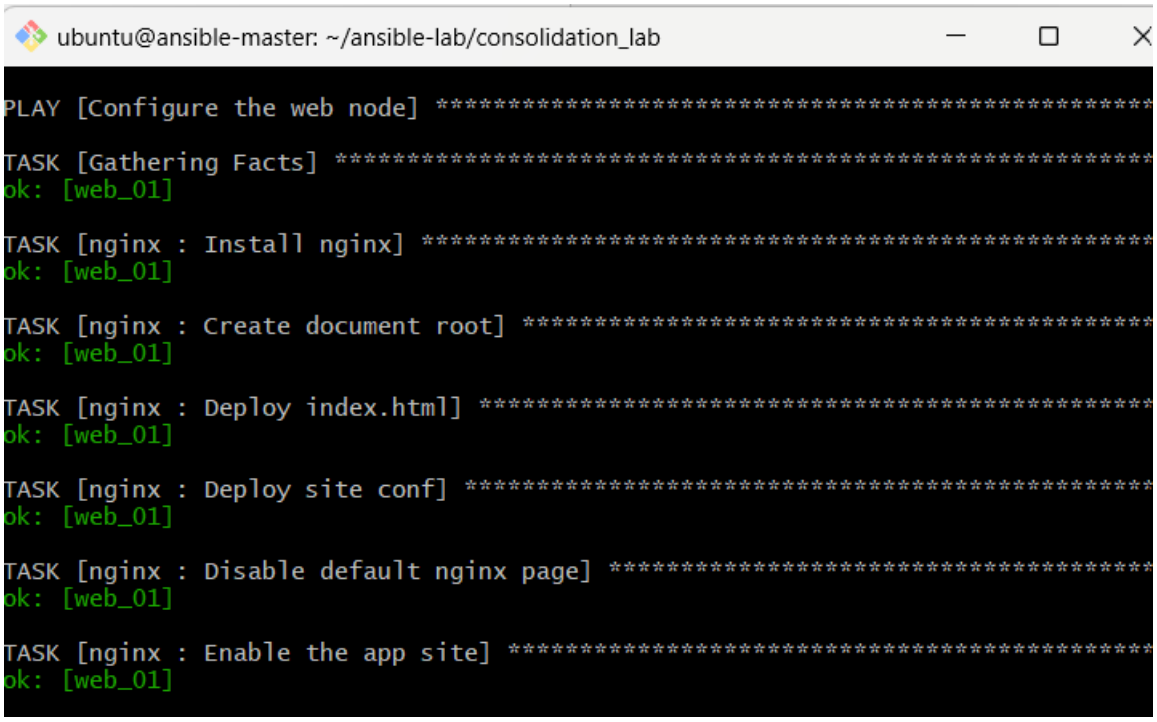


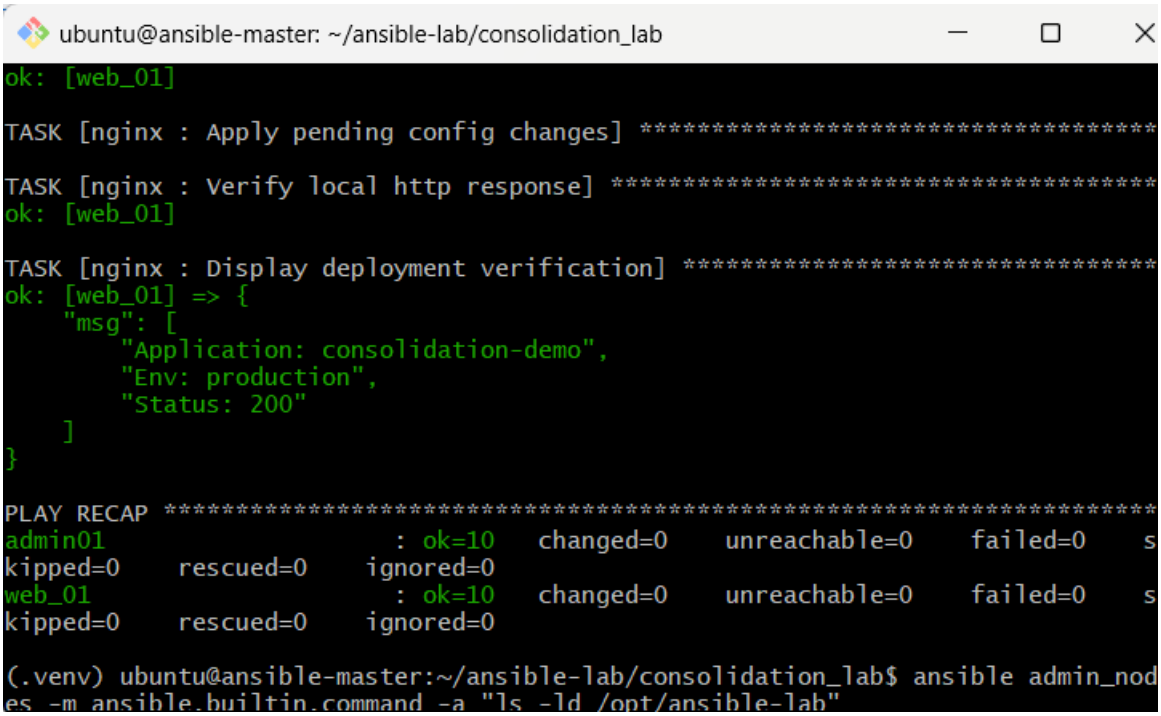
And there we go the outputs (both nodes combined):

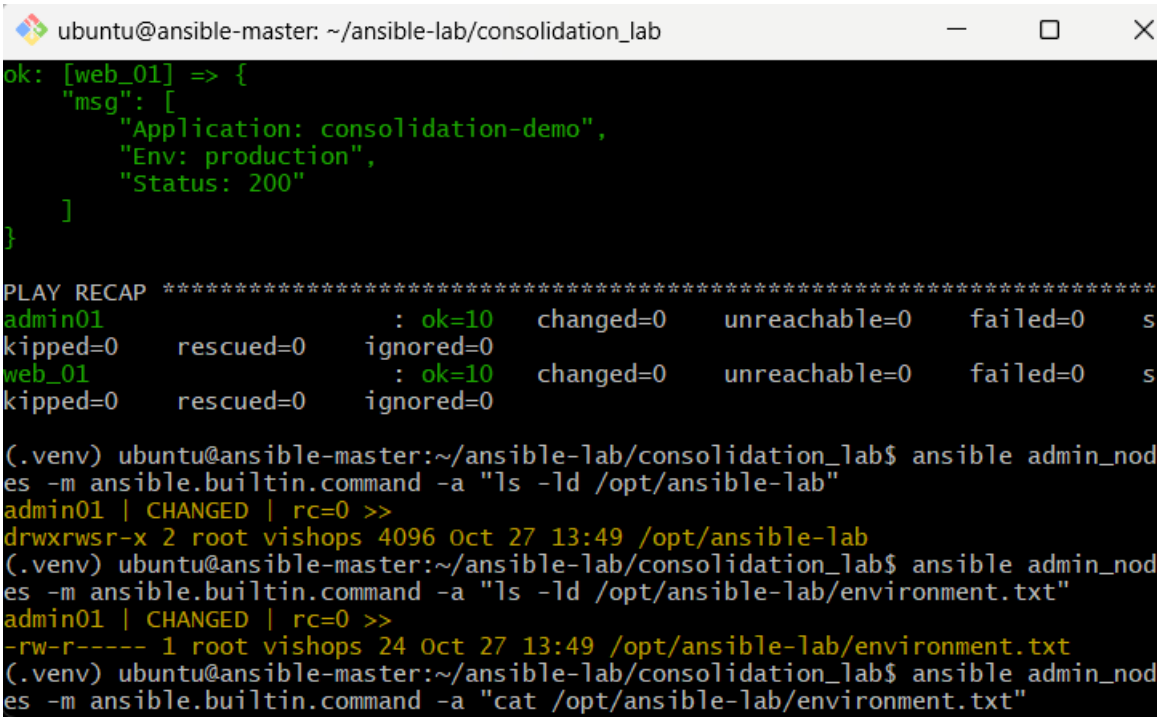


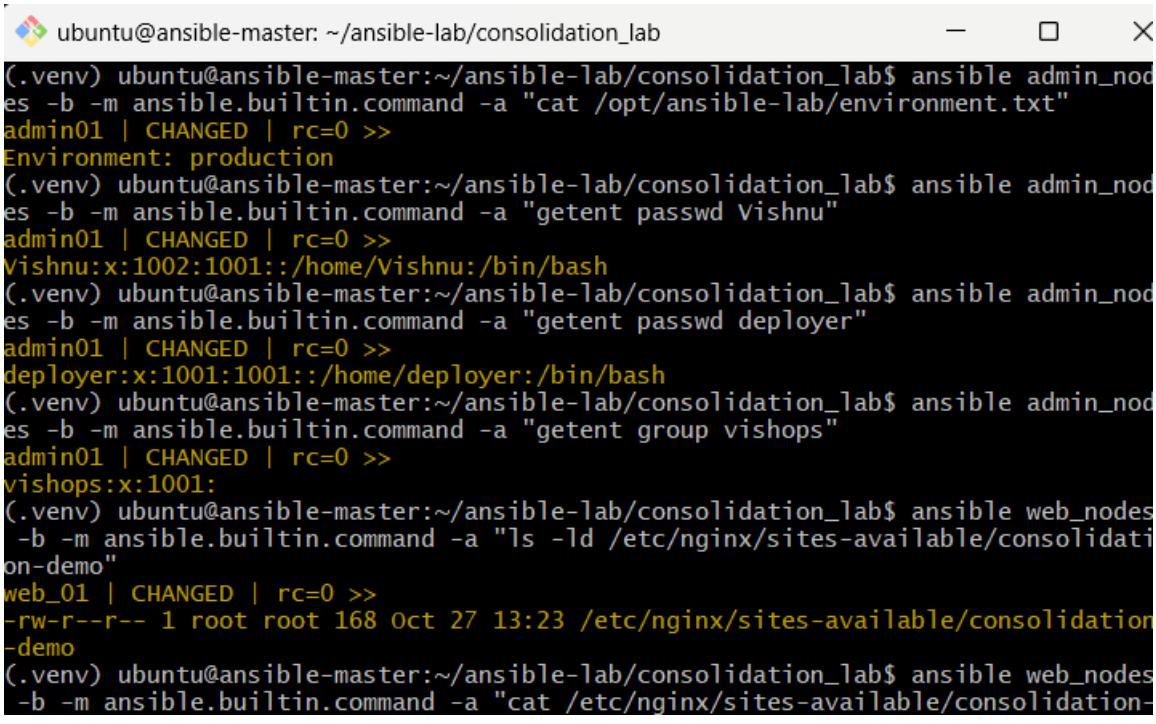

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
TASK [Gathering Facts] **
ok: [admin01]
TASK [admin : Creation of admin group] **
ok: [admin01]
TASK [admin : Creation of users] **
ok: [admin01] => (item={'name': 'deployer', 'shell': '/bin/bash'})
ok: [admin01] => (item={'name': 'Vishnu', 'shell': '/bin/bash'})
TASK [admin : Create the admin workspace] *************************************
ok: [admin01]
TASK [admin : Create an env specific marker] **********************************
ok: [admin01]
TASK [admin : Check if the env is set to production] **************************
ok: [admin01] => {
"msg": "Environment is set to Production"
}
TASK [admin : List the configured users] **************************************
ok: [admin01] => (item={'name': 'deployer', 'shell': '/bin/bash'}) => {

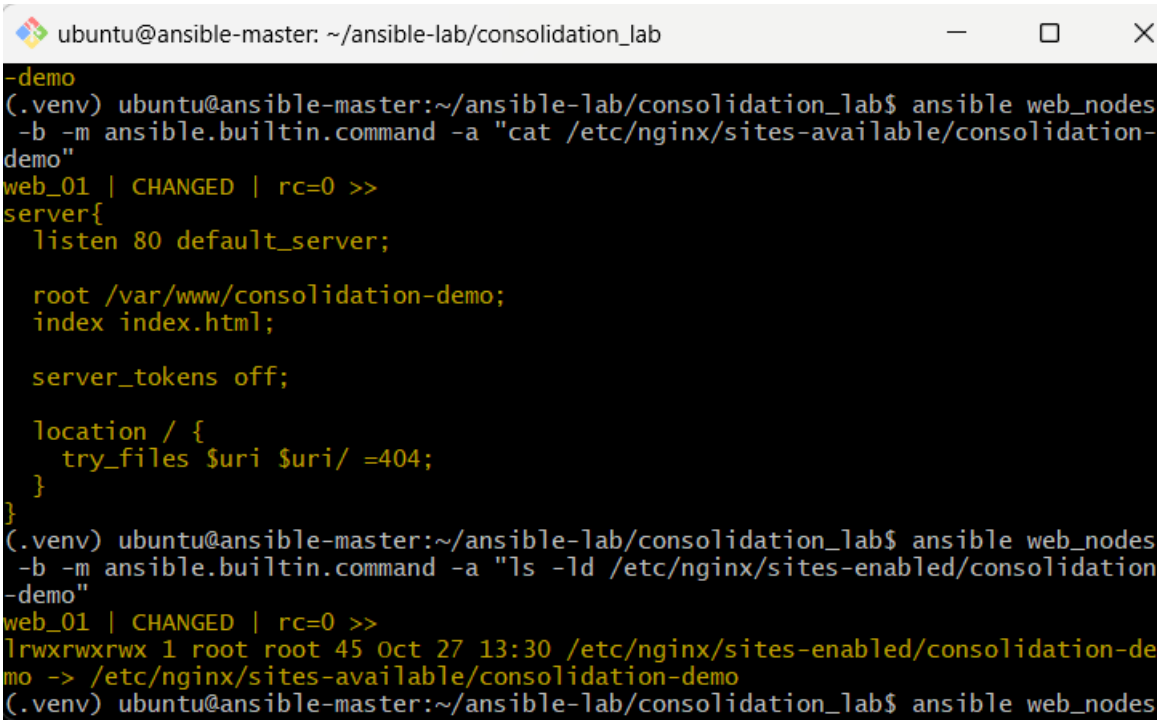

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
TASK [admin : List the configured users] **************************************
ok: [admin01] => (item={'name': 'deployer', 'shell': '/bin/bash'}) => {
"msg": "User deployer configured with shell /bin/bash"
}
ok: [admin01] => (item={'name': 'Vishnu', 'shell': '/bin/bash'}) => {
"msg": "User Vishnu configured with shell /bin/bash"
}
TASK [admin : Read a variable from the control node's env] ********************
ok: [admin01] => {
"msg": "Deployment label: local-lab"
}
TASK [admin : Run a variable with task-specific env vars] *********************
ok: [admin01]
TASK [admin : Display the managed node environment variable] ******************
ok: [admin01] => {
"msg": "production"
}
PLAY [Configure the web node] ***
TASK [Gathering Facts] **

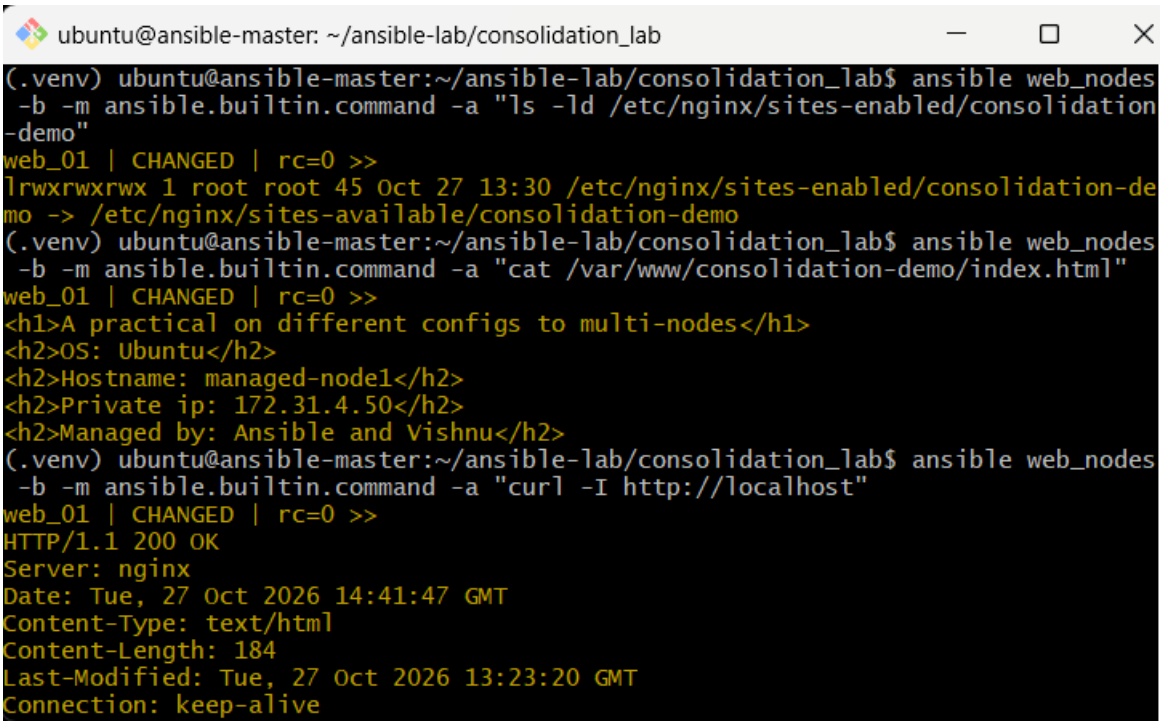

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
PLAY [Configure the web node] ***
TASK [Gathering Facts] **
ok: [web_01]
TASK [nginx : Install nginx] **
ok: [web_01]
TASK [nginx : Create document root] ***
ok: [web_01]
TASK [nginx : Deploy index.html] **
ok: [web_01]
TASK [nginx : Deploy site conf] ***
ok: [web_01]
TASK [nginx : Disable default nginx page] ***********************************
ok: [web_01]
TASK [nginx : Enable the app site] **
ok: [web_01]

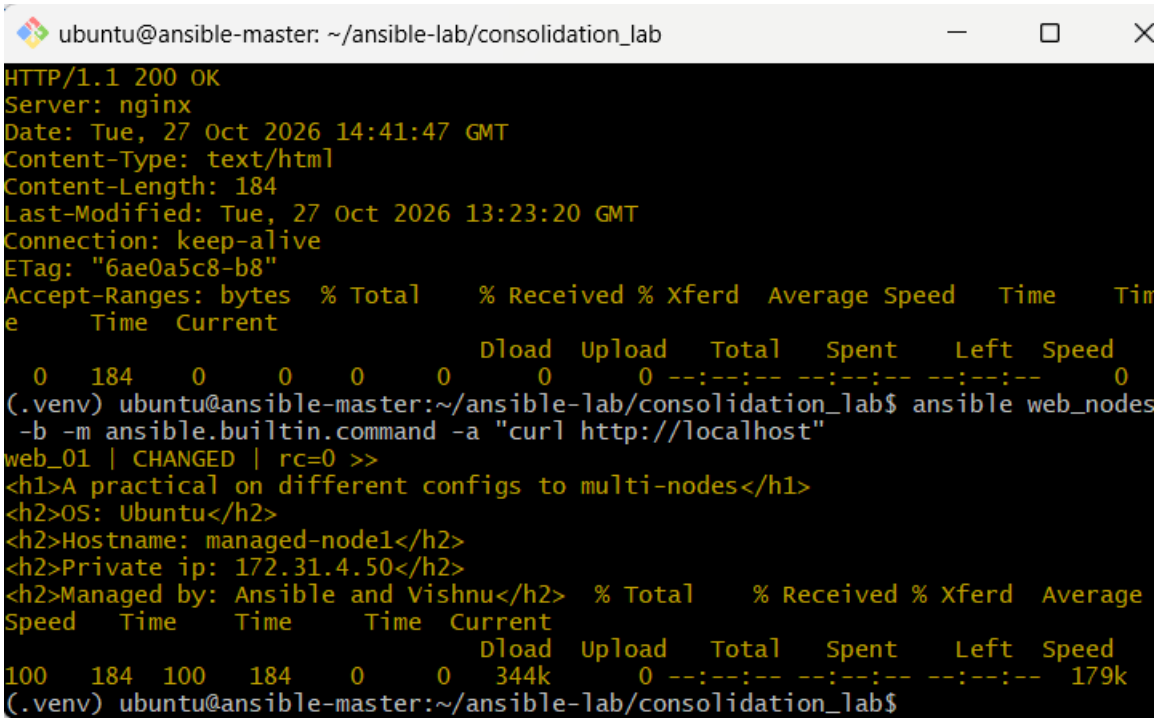

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
ok: [web_01]
TASK [nginx : Apply pending config changes] *********************************
TASK [nginx : Verify local http response] ***********************************
ok: [web_01]
TASK [nginx : Display deployment verification] ******************************
ok: [web_01] => {
"msg": [
"Application: consolidation-demo",
"Env: production",
"Status: 200"
]
}
PLAY RECAP **
admin01 : ok=10 changed=0 unreachable=0 failed=0 s
kipped=0 rescued=0 ignored=0
web_01 : ok=10 changed=0 unreachable=0 failed=0 s
kipped=0 rescued=0 ignored=0
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible admin_nod
es -m ansible.builtin.command -a "ls -ld /opt/ansible-lab"


ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
ok: [web_01] => {
"msg": [
"Application: consolidation-demo",
"Env: production",
"Status: 200"
]
}
PLAY RECAP
admin01 : ok=10 changed=0 unreachable=0 failed=0 s
kipped=0 rescued=0 ignored=0
web_01 : ok=10 changed=0 unreachable=0 failed=0 s
kipped=0 rescued=0 ignored=0
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible admin_nod
es -m ansible.builtin.command -a "ls -ld /opt/ansible-lab"
admin01 | CHANGED | rc=0 >>
drwxrwsr-x 2 root vishops 4096 Oct 27 13:49 /opt/ansible-lab
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible admin_nod
es -m ansible.builtin.command -a "ls -ld /opt/ansible-lab/environment.txt"
admin01 | CHANGED | rc=0 >>
-rw-r----- 1 root vishops 24 Oct 27 13:49 /opt/ansible-lab/environment.txt
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible admin_nod
es -m ansible.builtin.command -a "cat /opt/ansible-lab/environment.txt"

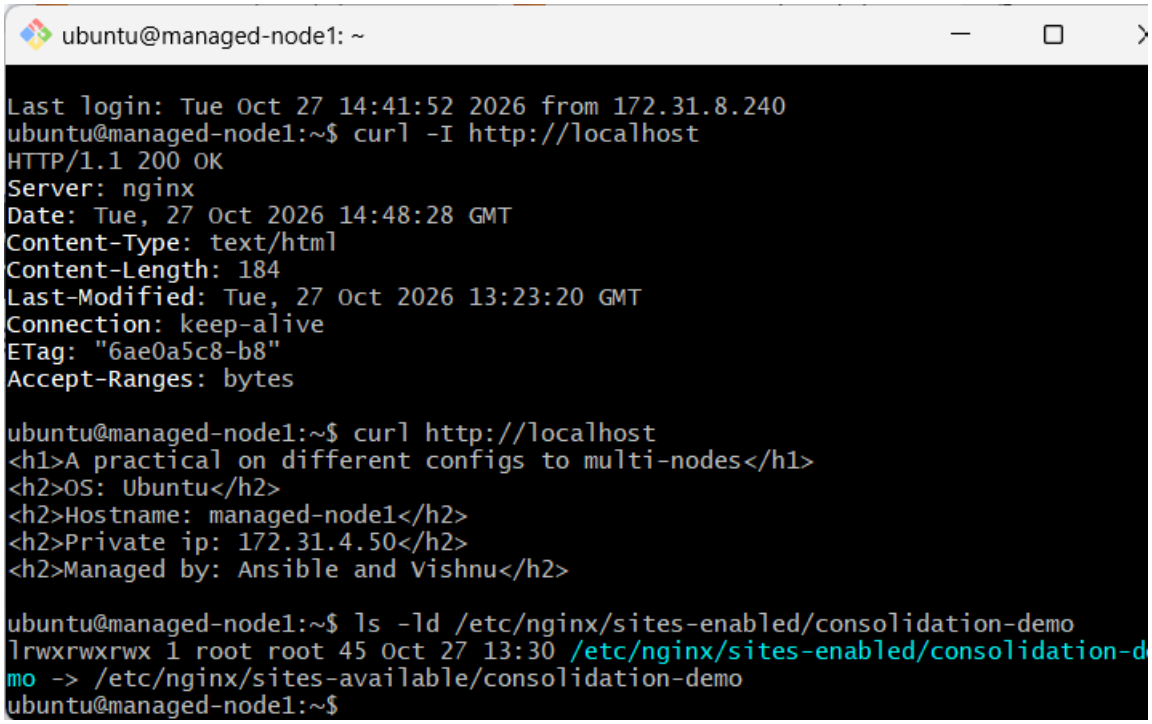

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible admin_nod
es -b -m ansible.builtin.command -a "cat /opt/ansible-lab/environment.txt"
admin01 | CHANGED | rc=0 >>
Environment: production
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible admin_nod
es -b -m ansible.builtin.command -a "getent passwd Vishnu"
admin01 | CHANGED | rc=0 >>
Vishnu:x:1002:1001::/home/Vishnu:/bin/bash
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible admin_nod
es -b -m ansible.builtin.command -a "getent passwd deployer"
admin01 | CHANGED | rc=0 >>
deployer:x:1001:1001::/home/deployer:/bin/bash
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible admin_nod
es -b -m ansible.builtin.command -a "getent group vishops"
admin01 | CHANGED | rc=0 >>
vishops:x:1001:
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible web_nodes
-b -m ansible.builtin.command -a "ls -ld /etc/nginx/sites-available/consolidati
on-demo"
web_01 | CHANGED | rc=0 >>
-rw-r--r-- 1 root root 168 Oct 27 13:23 /etc/nginx/sites-available/consolidation
-demo
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible web_nodes
-b -m ansible.builtin.command -a "cat /etc/nginx/sites-available/consolidation-

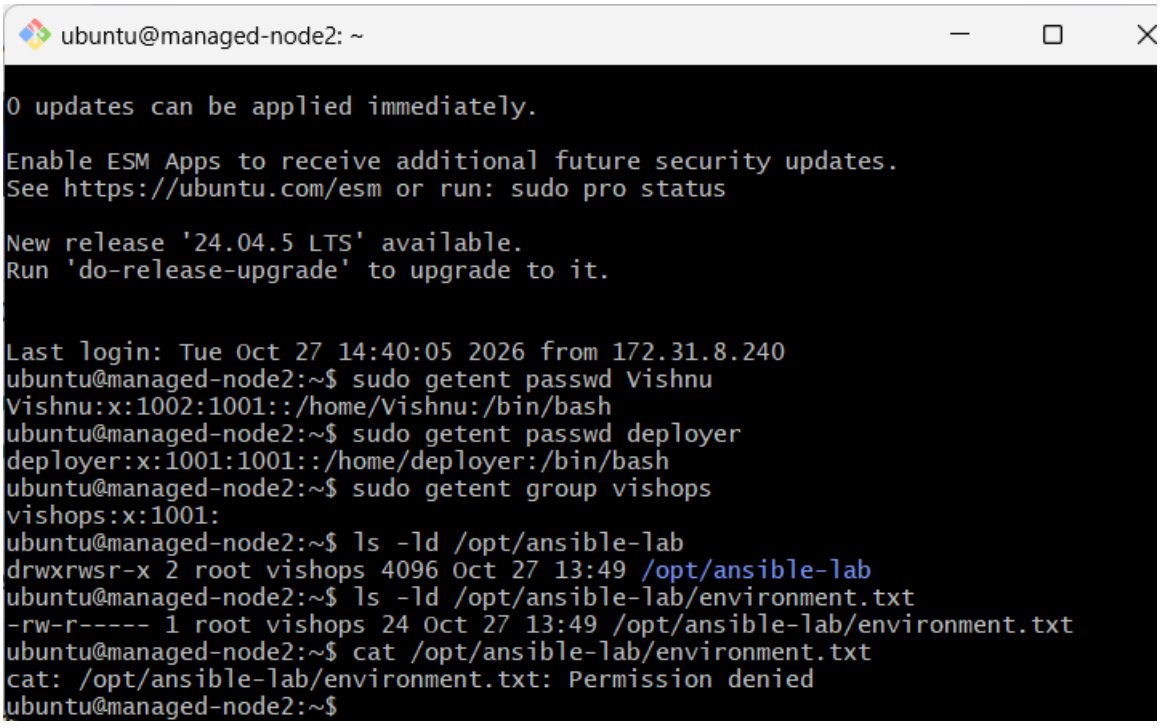

ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
-demo
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible web_nodes
 -b -m ansible.builtin.command -a "cat /etc/nginx/sites-available/consolidation-
demo"
web_01 | CHANGED | rc=0 >>
server{
  listen 80 default_server;
  root /var/www/consolidation-demo;
  index index.html;
  server_tokens off;
  location / {
    try_files $uri $uri/ =404;
  }
}
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible web_nodes
 -b -m ansible.builtin.command -a "ls -ld /etc/nginx/sites-enabled/consolidation
-demo"
web_01 | CHANGED | rc=0 >>
lrwxrwxrwx 1 root root 45 Oct 27 13:30 /etc/nginx/sites-enabled/consolidation-de
mo -> /etc/nginx/sites-available/consolidation-demo
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible web_nodes


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(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible web_nodes
 -b -m ansible.builtin.command -a "cat /var/www/consolidation-demo/index.html"
web_01 | CHANGED | rc=0 >>
<h1>A practical on different configs to multi-nodes</h1>
<h2>OS: Ubuntu</h2>
<h2>Hostname: managed-node1</h2>
<h2>Private ip: 172.31.4.50</h2>
<h2>Managed by: Ansible and Vishnu</h2>
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible web_nodes
 -b -m ansible.builtin.command -a "curl -I http://localhost"
web_01 | CHANGED | rc=0 >>
HTTP/1.1 200 OK
Server: nginx
Date: Tue, 27 Oct 2026 14:41:47 GMT
Content-Type: text/html
Content-Length: 184
Last-Modified: Tue, 27 Oct 2026 13:23:20 GMT
Connection: keep-alive


ubuntu@ansible-master: ~/ansible-lab/consolidation_lab
HTTP/1.1 200 OK
Server: nginx
Date: Tue, 27 Oct 2026 14:41:47 GMT
Content-Type: text/html
Content-Length: 184
Last-Modified: Tue, 27 Oct 2026 13:23:20 GMT
Connection: keep-alive
ETag: "6ae0a5c8-b8"
Accept-Ranges: bytes
% Total % Received % Xferd Average Speed Time Time
Time Current
Dload Upload Total Spent Left Speed
0 184 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$ ansible web_nodes
-b -m ansible.builtin.command -a "curl http://localhost"
web_01 | CHANGED | rc=0 >>
<h1>A practical on different configs to multi-nodes</h1>
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% Total % Received % Xferd Average
Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 184 100 184 0 0 344k 0 --:--:-- --:--:-- --:--:-- 179k
(.venv) ubuntu@ansible-master:~/ansible-lab/consolidation_lab$


65.1.107.184
Not secure
65.1.107.184
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmarks
A practical on different configs to multi-nodes
OS: Ubuntu
Hostname: managed-node1
Private ip: 172.31.4.50
Managed by: Ansible and Vishnu


ubuntu@managed-node1: ~
Last login: Tue Oct 27 14:41:52 2026 from 172.31.8.240
ubuntu@managed-node1:~$ curl -I http://localhost
HTTP/1.1 200 OK
Server: nginx
Date: Tue, 27 Oct 2026 14:48:28 GMT
Content-Type: text/html
Content-Length: 184
Last-Modified: Tue, 27 Oct 2026 13:23:20 GMT
Connection: keep-alive
ETag: "6ae0a5c8-b8"
Accept-Ranges: bytes
ubuntu@managed-node1:~$ curl http://localhost
<h1>A practical on different configs to multi-nodes</h1>
<h2>OS: Ubuntu</h2>
<h2>Hostname: managed-node1</h2>
<h2>Private ip: 172.31.4.50</h2>
<h2>Managed by: Ansible and Vishnu</h2>
ubuntu@managed-node1:~$ ls -ld /etc/nginx/sites-enabled/consolidation-demo
lrwxrwxrwx 1 root root 45 Oct 27 13:30 /etc/nginx/sites-enabled/consolidation-d
mo -> /etc/nginx/sites-available/consolidation-demo
ubuntu@managed-node1:~$


ubuntu@managed-node2: ~
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
Last login: Tue Oct 27 14:40:05 2026 from 172.31.8.240
ubuntu@managed-node2:~$ sudo getent passwd Vishnu
Vishnu:x:1002:1001::/home/Vishnu:/bin/bash
ubuntu@managed-node2:~$ sudo getent passwd deployer
deployer:x:1001:1001::/home/deployer:/bin/bash
ubuntu@managed-node2:~$ sudo getent group vishops
vishops:x:1001:
ubuntu@managed-node2:~$ ls -ld /opt/ansible-lab
drwxrwsr-x 2 root vishops 4096 Oct 27 13:49 /opt/ansible-lab
ubuntu@managed-node2:~$ ls -ld /opt/ansible-lab/environment.txt
-rw-r----- 1 root vishops 24 Oct 27 13:49 /opt/ansible-lab/environment.txt
ubuntu@managed-node2:~$ cat /opt/ansible-lab/environment.txt
cat: /opt/ansible-lab/environment.txt: Permission denied
ubuntu@managed-node2:~$